

Daniel Scalena

📍 Italy | Netherlands ✉ scalena99@gmail.com ☎ +39 3318397458 🌐 danielsc4.it

Education

Double Doctorate (Ph.D) in Computer Science - NLP Nov 2023 – exp. Q4 2026

University of Milano - Bicocca | University of Groningen (CLCG)

Researching interpretability to improve the safety, reliability, and real-world applicability of generative models.

Mediterranean Machine Learning (M2L) Summer School Sep 2024

Milano, Italy

1-week full-time lectures and practical sessions led primarily by Google DeepMind scientists; presented my [paper](#).

M.Sc in Computer Science, ML track Oct 2021 – Oct 2023

University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 29.7/30)
- Thesis: On the explainability of Large Language Models detoxification.

B.Sc in Computer Science Oct 2018 – Jul 2021

University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 28.3/30)
- Thesis: Hate speech detection on social networks using state-of-the-art NLP techniques.

Experience

University of Groningen, Computational Linguistic Group Apr 2023 – Jul 2023

Research Intern

- Aligned LMs to counter hate speech narratives using RLAIIF, reaching up to 50% in safety improvements;
- Analyzed and interpreted model behavior shifts after applying post-training algorithms.

C.I.N.I. & University of Milano - Bicocca Jun 2022 - Apr 2023

Assistant Researcher

- Devised Open Relation Extraction for Italian legal texts, improving previous accuracy by 30%;
- Reduced manual legal document review using ad hoc models;
- Project funded by the *Italian Ministry of Justice* and *Inter-university Consortium for Computer Science*.

TESTUDO s.r.l. Jun 2022 - Aug 2022

ML Engineer (contractor)

- Designed and trained deep learning models to predict working hours timestamps;
- Fully automated the company's recognition system.

Main publications

🔗 **EAGer: Entropy-Aware GEneration for Adaptive Inference-Time Scaling**

Daniel Scalena, Leonidas Zotos, Elisabetta Fersini, Malvina Nissim, Ahmet Üstün

Proceedings of the 43rd International Conference on Machine Learning, (ICML 2026)

🔗 **Steering Large Language Models for Machine Translation Personalization**

Daniel Scalena*, Gabriele Sarti*, Arianna Bisazza, Malvina Nissim, Elisabetta Fersini

Proceedings of the 19th Conference of the European Chapter of the Association for CL, (EACL 2026)

🔗 **Multi-property steering of large language models with dynamic activation composition**

Daniel Scalena, Gabriele Sarti, Malvina Nissim

Proceedings at the 7th BlackBoxNLP Workshop, (at EMNLP 2024)

[🔗](#) **A gentle push funziona benissimo: making instructed models in Italian via contrastive activation steering**

Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini

Proceedings at the 10th Italian Conference on Computational Linguistics, (CLIC-it 2024)

[🔗](#) **Let the Models Respond: Interpreting Language Model Detoxification Through the Lens of Prompt Dependence**

Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini

Extended abstract at the 6th BlackboxNLP Workshop, (at EMNLP 2023)

[Full list of publications](#) [🔗](#)

Scholarships and Awards

Google and Apple - BlackBoxNLP (EMNLP 2023) travel award	<i>2023</i>
Full-Fee M.Sc. Scholarship due to merit and financial scores	<i>2021 - 2023</i>
Full-Fee B.Sc. Scholarship due to merit and financial scores	<i>2018 - 2021</i>

Projects

Reward LM	<i>DanielSc4/RewardLM</i> 🔗
- Python toolkit library for Fine-Tuning and Reinforcement Learning from AI Feedback of Generative LMs; - Developed tools to interpret and compare toxicity differences between original and fine-tuned models.	

Activities

Co-organisier	Challenge the Abilities of LAnguage Models in ITALian (CALAMITA) <i>2024</i>
Teaching	NLP course labs, MSc Data Science @ UniMiB (<i>2024 - 2025</i>)
Reviewer	NeurIPS <i>2026</i> , ICML <i>2026</i> , EMNLP <i>2025</i> , BlackBox-NLP <i>2025</i> , ACL <i>2025</i> , LREC-Coling <i>2024</i> , NAACL <i>2024*</i> , Information Processing and Management <i>2024 - 2025</i> , CliC-IT <i>2023 - 2024</i> , ICML Mechanistic Interpretability Workshop <i>2024*</i> (*secondary reviewer)
Volunteer	CliC-IT <i>2021</i> , EMNLP <i>2023</i>
Misc.	European Researchers Night GroNLP collaborator <i>2023</i>

Languages

Italian	Native Language
English	C1 - B-between Foreign Languages 🔗
French	B1